

Exercise sheet 3 : correspondences

1. Here is a table that shows courses chosen by 6 students, followed by a list of the assigned instructors.

	Analysis	Physics	Chemistry	Programming	Calculus	Phys. Lab	Chem. Lab
Charles	✓	✓		✓			
Charlie		✓		✓	✓		
Carlo			✓		✓		✓
Karl	✓	✓				✓	
Charuzu		✓	✓		✓		
Charlot	✓	✓	✓				

- Analysis : Maicon,
- Physics : Mikkel,
- Chemistry : Michelle,
- Programming : Mikhail,
- Calculus : Micheal,
- Physics Lab : Mitchell,
- Chemistry Lab : Mihai.

Let X denote the set that consists of these 6 students, Y denote the set of all the courses, Z denote the set of all the instructors, f denote the correspondence from X to Y based on the above table, g denote the correspondence from Y to Z based on the assignment of the course instructors.

- (1) Draw a table of g .
- (2) Is the correspondence f injective ? surjective ? a function ? a multi-valued mapping ? What about g ?
- (3) Draw a table of the composite $g \circ f$. What can we conclude from it ?

- (4) Draw a table of the composite $f^{-1} \circ f$. What can we conclude from it?
2. For each of the following mappings, determine if it is injective, surjective. Justify your answer.
- (1) $f : \mathbb{Z} \rightarrow \mathbb{Z}, f(x) = 3x + 4$.
 - (2) $f : \mathbb{N} \rightarrow \mathbb{N}, f(x) = 3x + 4$.
 - (3) $f : \mathbb{Z} \rightarrow \mathbb{Z}, f(x) = 2x + 4$.
 - (4) $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2, f(x, y) = (x + y, x - y)$.
3. For each of the following functions from $\mathbb{R}$ to $\mathbb{R}$, determine its domain of definition and its range.
- (1) $\cos \circ \sin$.
 - (2) $\exp \circ \ln$.
 - (3) $\ln \circ \exp$.
 - (4) ι defined as $\iota(x) = x^{-1}$.
 - (5) $\iota \circ \iota$.
4. Consider the correspondence φ from $\mathbb{R}$ to $\mathbb{R}$ whose graph is given by

$$\{(x, y) : x^2 + y^2 \leq 1\}.$$

- (1) Draw the graph of the correspondence φ .
 - (2) Determine the domain of definition and the range of φ .
 - (3) We view φ as a binary relation on $\mathbb{R}$. Is this binary relation reflexive, symmetric, transitive?
 - (4) Draw the graph of $\varphi \circ \varphi$.
5. Consider the mapping $f : \mathbb{R} \rightarrow \mathbb{R}$ defined as $f(x) = x^2$. Let

$$A = \{x \in \mathbb{R} \mid -1 \leq x \leq 4\}.$$

Determine $f(A)$ and $f^{-1}(A)$.

6. Let f and g be functions from $\mathbb{R}$ to $\mathbb{R}$ defined as

$$f(x) = 3x + 1, \quad g(x) = x^2 - 1.$$

Determine the functions $f \circ g$ and $g \circ f$. Are they equal?

7. For each of the following function h from $\mathbb{R}$ to $\mathbb{R}$. Determine two functions f and g such that $h = f \circ g$.
- (1) $h(x) = \sqrt{3x - 1}$,

$$(2) \ h(x) = \sin(x + \pi/2),$$

$$(3) \ h(x) = 1/(x + 1).$$

8. Let f and g be mappings from $\mathbb{N}$ to $\mathbb{N}$ defined as

$$f(n) = 2n, \quad g(n) = \begin{cases} n/2, & \text{if } n \text{ is even,} \\ 0, & \text{if } n \text{ is odd.} \end{cases}$$

Determine $g \circ f$ and $f \circ g$. Is the mapping f injective, surjective, a bijection? Is the mapping g injective, surjective, a bijection?

9. Let $f : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$ be the mapping sending $(n, p) \in \mathbb{N} \times \mathbb{N}$ to $2^n(2p+1)$.

(1) Show that f is injective.

(2) Determine the range of f .

10. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a mapping. For each of the following statements, construct a statement of the opposite truth value, where no quantifier is preceded by a negation symbol.

(1) $\forall x \in \mathbb{R}, f(x) \neq 0$,

(2) $\forall M \in \mathbb{R}_{>0}, \exists A \in \mathbb{R}_{>0}, \forall x \geq A, f(x) > M$.

(3) $\forall x \in \mathbb{R}, f(x) > 0 \Rightarrow x \leq 0$.

(4) $\forall \varepsilon \in \mathbb{R}_{>0}, \exists \eta \in \mathbb{R}_{>0}, \forall (x, y) \in \mathbb{R} \times \mathbb{R}, |x-y| < \eta \Rightarrow |f(x)-f(y)| < \varepsilon$.

11. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a mapping. By using the quantifiers, express the following statements :

(1) f is constant.

(2) 0 lies in the range of f .

(3) f only takes non-negative values.

12. We consider the following condition $P(\cdot)$ on the set $\mathbb{R}^{\mathbb{R}}$ of mappings from $\mathbb{R}$ to $\mathbb{R}$

$$P(f) := “\exists M \in \mathbb{R}, \forall x \in \mathbb{R}, f(x) \leq M”.$$

(1) Construct a condition $Q(\cdot)$ on $\mathbb{R}^{\mathbb{R}}$ without quantifier preceded by a negation symbol, such that, for any $f \in \mathbb{R}^{\mathbb{R}}$, $Q(f)$ and $\neg P(f)$ have the same truth value.

(2) Construct an example of mapping in $\mathbb{R}^{\mathbb{R}}$ that satisfies the condition $P(\cdot)$.

(3) Construct an example of mapping in $\mathbb{R}^{\mathbb{R}}$ that does not satisfies the condition $P(\cdot)$.

- (4) For each of the following conditions, determine if it is satisfied by all mappings in $\mathbb{R}^{\mathbb{R}}$, if it is not satisfied by none of the mappings in $\mathbb{R}^{\mathbb{R}}$, or if it is satisfied by some but not all of the mappings in $\mathbb{R}^{\mathbb{R}}$.
- a) $\exists x \in \mathbb{R}, \forall M \in \mathbb{R}, f(M) \leq x$.
 - b) $\exists x \in \mathbb{R}, \forall M \in \mathbb{R}, f(x) \leq M$.
 - c) $\forall M \in \mathbb{R}, \exists x \in \mathbb{R}, f(x) \leq M$.
 - d) $\forall M \in \mathbb{R}, \exists x \in \mathbb{R}, f(M) \leq x$.
- 13.** Let n and p be two natural numbers, $X = \{1, \dots, n\}$ and $Y = \{1, \dots, p\}$.
- (1) How many correspondences are there from X to Y ?
 - (2) How many injective correspondences are there from X to Y ?
 - (3) How many surjective correspondences are there from X to Y ?
 - (4) How many functions are there from X to Y ?
 - (5) How many mappings are there from X to Y ?
 - (6) How many bijections are there from X to Y .
- 14.** For any natural number n , let E_n be the set $\{1, \dots, n\}$ (by convention E_0 denotes the empty set). For any $(n, p) \in \mathbb{N}^2$, let $S_{n,p}$ be the number of surjective mappings from E_n to E_p .
- (1) Determine $S_{n,p}$ for $(n, p) \in \mathbb{N}^2$ such that $p > n$.
 - (2) For any $n \in \mathbb{N}$, determine $S_{n,0}$.
 - (3) Show that, for any non-zero natural number p , one has

$$\sum_{i=0}^p (-1)^i \binom{p}{i} = 0,$$

where

$$\binom{n}{p} := \frac{n!}{p!(n-p)!}$$

- (4) Let i, q and p be natural numbers such that $i \leq q \leq p$. Show that

$$\binom{p}{q} \binom{q}{i} = \binom{p}{i} \binom{p-i}{q-i}.$$

- (5) Show that, for any $(i, p) \in \mathbb{N}^2$ such that $i < p$, one has

$$\sum_{q=i}^p (-1)^q \binom{p}{q} \binom{q}{i} = 0.$$

- (6) Let p and q be natural numbers such that $q \leq p$. Show that the number of mappings from E_n to E_p whose range has exactly q elements is equal to $S_{n,q} \binom{p}{q}$.
- (7) Let p and n be natural numbers such that $p \leq n$ and $n \geq 1$. Show that

$$p^n = \sum_{q=1}^p \binom{p}{q} S_{n,q}.$$

- (8) Let p and n be natural numbers such that $p \leq n$ and $n \geq 1$. Show that

$$(-1)^p S_{n,p} = \sum_{i=1}^p (-1)^i \binom{p}{i} i^n.$$

- (9) Let p and n be natural numbers such that $1 \leq p < n$. Show that

$$S_{n,p} = p(S_{n-1,p} + S_{n-1,p-1}).$$

- (10) Let $p \in \mathbb{N}$. Determine $S_{p,p}$ and $S_{p+1,p}$.
- (11) Let p and n be natural numbers such that $p \leq n$. Show that the number of surjective functions from E_n to E_p is equal to

$$\sum_{j=p}^n \binom{n}{j} S_{j,p}.$$

- (12) Prove the equality

$$\sum_{j=p}^n \binom{n}{j} S_{j,p} = \frac{1}{p+1} S_{n+1,p+1}.$$

- 15.** For any natural number n , let E_n be the set $\{1, \dots, n\}$ (by convention E_0 denotes the empty set).

- (1) Let n and p be natural numbers such that $n \leq p$. Show that the number of injective mappings from E_n to E_p is equal to $p!/(p-n)!$.
- (2) Show that the number of injective functions from E_n to E_p is

$$\sum_{k=0}^{\min\{n,p\}} \binom{n}{k} \binom{p}{k} k!.$$

- 16.** Let X be a set and $\mathcal{P}(X)$ be the power set of X , namely $\mathcal{P}(X)$ is the set of all subsets of X . The purpose of this exercise is to prove that there does not exist any injective mapping from $\mathcal{P}(X)$ to X . We reason by contradiction assuming that $\Phi : \mathcal{P}(X) \rightarrow X$ is an injective mapping.

- (1) Show that Φ^{-1} is a function from X to $\mathcal{P}(X)$.
- (2) Let A be the set of $n \in \text{Im}(\Phi)$ such that $n \notin \Phi^{-1}(n)$. Show that $\Phi^{-1}(\Phi(A)) = A$.
- (3) Examine the truth value of the statement $\Phi(A) \in A$ and obtain a contradiction.